



MCS3301 Introduction to Matlab for Control

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Differential Equations

- `dsolve` can be used to solve DEs after declaring the time dependent variable as **symbolic**.
- For example, solve the equation

$$\frac{dv}{dt} + 5v = 10$$

```
>> syms v(t)
>> de = diff(v,t)+5*v == 10
de(t) =
D(v)(t) + 5*v(t) == 10
>> dsolve(de)
ans =
C5*exp(-5*t) + 2
```

- To give initial conditions

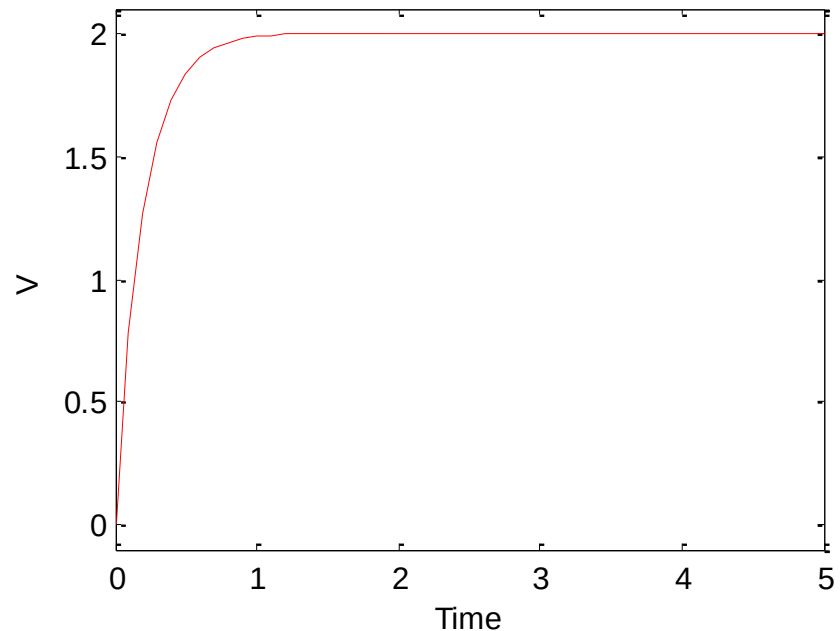
```
>> ic = v(0) == 1;
>> dsolve(de, ic)
ans =
2 - exp(-5*t)
>> ic = v(0) == 0;
>> dsolve(de, ic)
ans =
2 - 2*exp(-5*t)
```



Differential Equations

- One (crude) method of plotting the DE is to generate values for the axes directly.

```
syms v(t)
de = diff(v,t)+5*v == 10;
ic = v(0) == 0;
dsolve(de,ic);
t = 0:0.1:10;
y = 2-2*exp(-5*t);
plot(t,y,'r')
xlabel('Time')
ylabel('V')
ylim([-0.1 2.1])
xlim([0 5])
```





Laplace Transforms

- Given the function $f(t) = e^{-at}$
- Its Laplace transform can be obtained in Matlab using the `laplace` statement.

```
>> syms a t
>> f = exp(-a*t);
>> F = laplace(f)
F =
1/(a + s)
```

- The `ilaplace` function generates the inverse Laplace transform.

```
>> ilaplace(F)
ans =
exp(-a*t)
```



Transfer Functions

- Transfer functions can be obtained in Matlab using the `tf` command.

- Given the first order transfer function $G(s) = \frac{10}{s+5}$

```
>> G = tf(10,[1 5])
```

```
G =
```

```
10
```

```
-----
```

```
s + 5
```

- The `step` keyword can be used to obtain the output response to a step input.

```
>> step(G)
```

